Cell Biology

Dr. Reshna K. Albarzanji
B.Sc. M.Sc. PhD.
Cytogenetics
Hawler Medical University
College of Medicine
Microbiology dep.
reshna.kamal@hmu.edu.krd

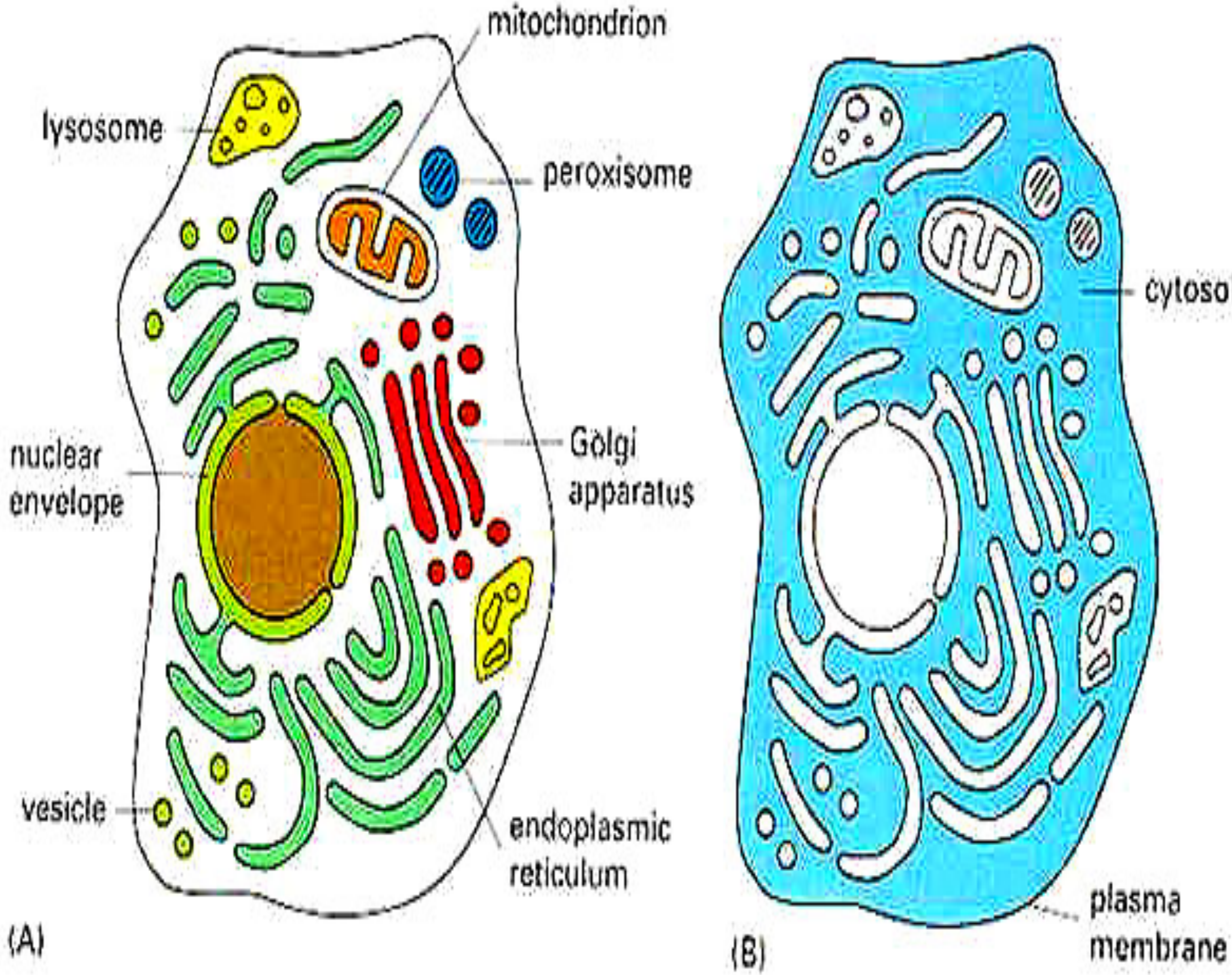
SGL : 4

Structure ,Position and function of cell organelles

Cytoplasm

The cytoplasm is composed of matrix or cytosol in which several membrane bound components called organelles & inclusions, plus deposits of carbohydrate, lipids, and pigment are embedded in it.

- But studies with high –voltage E.M. show a very complex three dimensional structure network of thin microtrabecules which anchor most of the cytoplasmic organelles.


mitochondrion
lysosome
peroxisome
nuclear envelope
Golgi apparatus
vesicle
endoplasmic reticulum
(A)
cytosol
plasma membrane
(B)

The endomembrane system:

Many of the different membranes of the eukaryotic cell are parts of an endomembrane system.

These membranes are related either through direct physical continuity or by the transfer of membrane segments through the movement of tiny vesicles.

The endomembrane system:

These relationships, however, do not mean that the various membranes are alike in structure and function, the endomembrane system includes the nuclear envelope ,endoplasmic reticulum , Golgi apparatus, lysosomes, various kinds of vacuoles.

1-Endoplasmic reticulum (E.R.)

Introduction

- Cytoplasm of eukaryotic cells contains a net work of membranous tubules & sacs formed by a continuous membrane which encloses a space called (cisterna).
- the endoplasmic reticulum membrane separates its internal compartment (the cisternal space) from the cytosol.

- **Structure of the rough Endoplasmic Reticulum**
- Is a series of connected flattened sacs, part of a continuous membrane organelle within the cytoplasm, that plays a central role in the synthesis of proteins.
- The rough endoplasmic reticulum (RER) is so named for the appearance of its outer surface, which is studded with protein-synthesizing particles known as ribosomes.
- The RER membrane is continuous with the nuclear envelope.
- The RER is also located near the Golgi apparatus, Many proteins that are synthesized in the RER are packaged into vesicles and transported to the Golgi apparatus.

Functions of the RER

1.Proteins synthesized on ribosomes of RER.

2.Modification and processing of newly synthesized proteins ;

A. glycosylation in the RER.

B. the folding of proteins.

C. the formation of di-sulfide bonds within polypeptide.

3. Transport of the proteins .

4. Manufacture of lysosomal enzymes

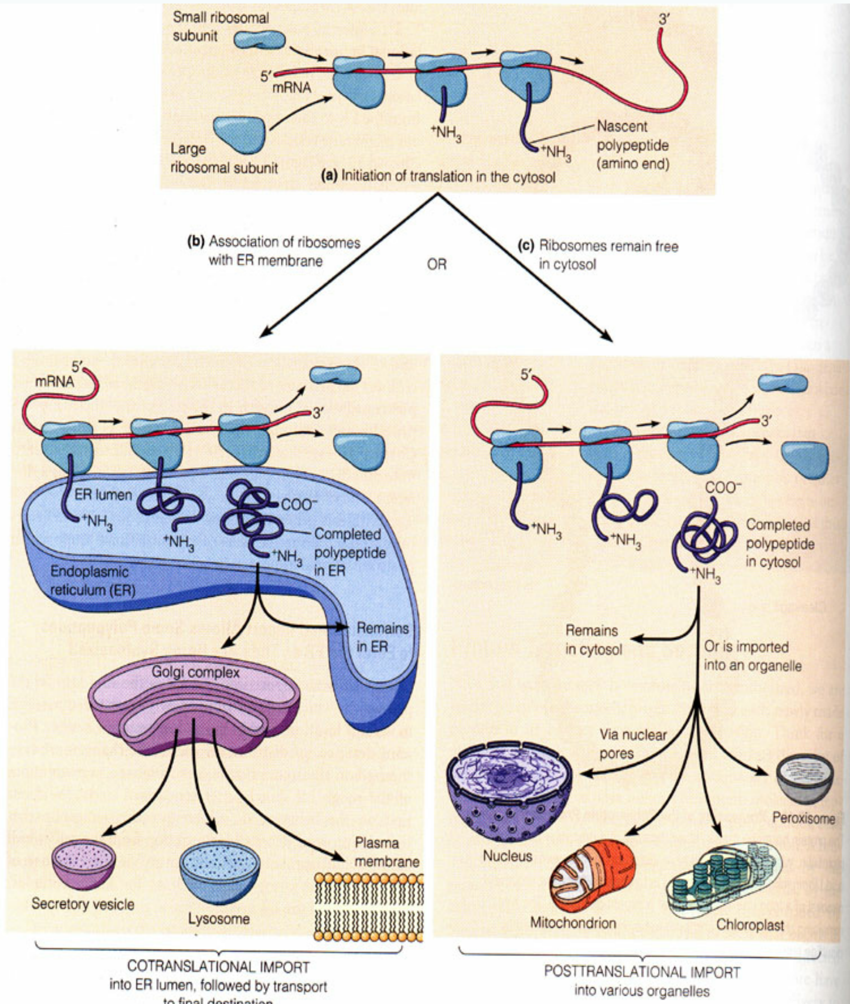


Figure 20-14 Intracellular Sorting of Proteins. **(a)** Synthesis of all polypeptides encoded by nuclear genes begins in the cytosol. The large and small ribosomal subunits associate with each other and with the 5′ end of an mRNA molecule, forming a functional ribosome that starts making the polypeptide. When the polypeptide is about 30 amino acids long, it enters one of two alternative pathways. **(b)** If destined for any of the compartments of the endomembrane system, the newly forming polypeptide becomes associated with the ER membrane and is transferred across the membrane into the lumen (cisternal space) of the ER as synthesis continues. This process is cotranslational import. The completed polypeptide then either remains in the ER or is transported via various vesicles and the Golgi complex to another final destination. (Integral membrane proteins are inserted into the ER membrane as they are made, rather than into the lumen.) **(c)** If the polypeptide is destined for the cytosol or for import into the nucleus, mitochondria, chloroplasts, or peroxisomes, its synthesis continues in the cytosol. When the polypeptide is complete, it is released from the ribosome and either remains in the cytosol or is transported into the appropriate organelle by posttranslational import. Polypeptide uptake by the nucleus occurs via the nuclear pores, using a mechanism different from that involved in posttranslational uptake by other organelles.

Proteins synthesized on ribosomes of rER include:

- **Secretory proteins;**
- **integral membrane proteins;**
- **soluble proteins of organelles.**

- **Smooth endoplasmic reticulum**
- The smooth endoplasmic reticulum is present in almost all eukaryotic cells. However, some eukaryotic cells lack sER, such as **ova, embryonic cells, and mature RBCs**. Conversely, some specialized cells, are rich in sER.
- Examples of cells with abundant sER are sebaceous glands, gonadal cells (the steroid-secreting cells, such as Leydig cells in the testis and follicular cells in the ovary) involved in producing steroid hormones and other hormones of the endocrine system, hepatocytes in the liver, and cells of striated muscles.

- **Smooth Endoplasmic Reticulum Functions**
- The major functions of sER include
- **1-lipid synthesis**.
- **2-carbohydrate metabolism**.
- 3- **regulation of intracellular calcium concentration.**
- 4- **drug detoxification.**
- sER contains a wide range of enzymes involved in the synthesis of lipids, especially the biosynthesis of phospholipids and steroids.

- **Lipid synthesis**
- The smooth endoplasmic reticulum is the major site for lipid synthesis, particularly at the *membrane contact sites* (MCS).
- MCS are areas where ER membranes make close contact with other cytoplasmic organelles, such as the Golgi apparatus, mitochondria ,lysosomes , peroxisomes, endosomes, and plasma membrane, and allow the transfer of substances.
- The endoplasmic reticulum (ER)-localized enzymes present in these sites of sER are responsible for the synthesis of the majority of cellular lipids.
- .

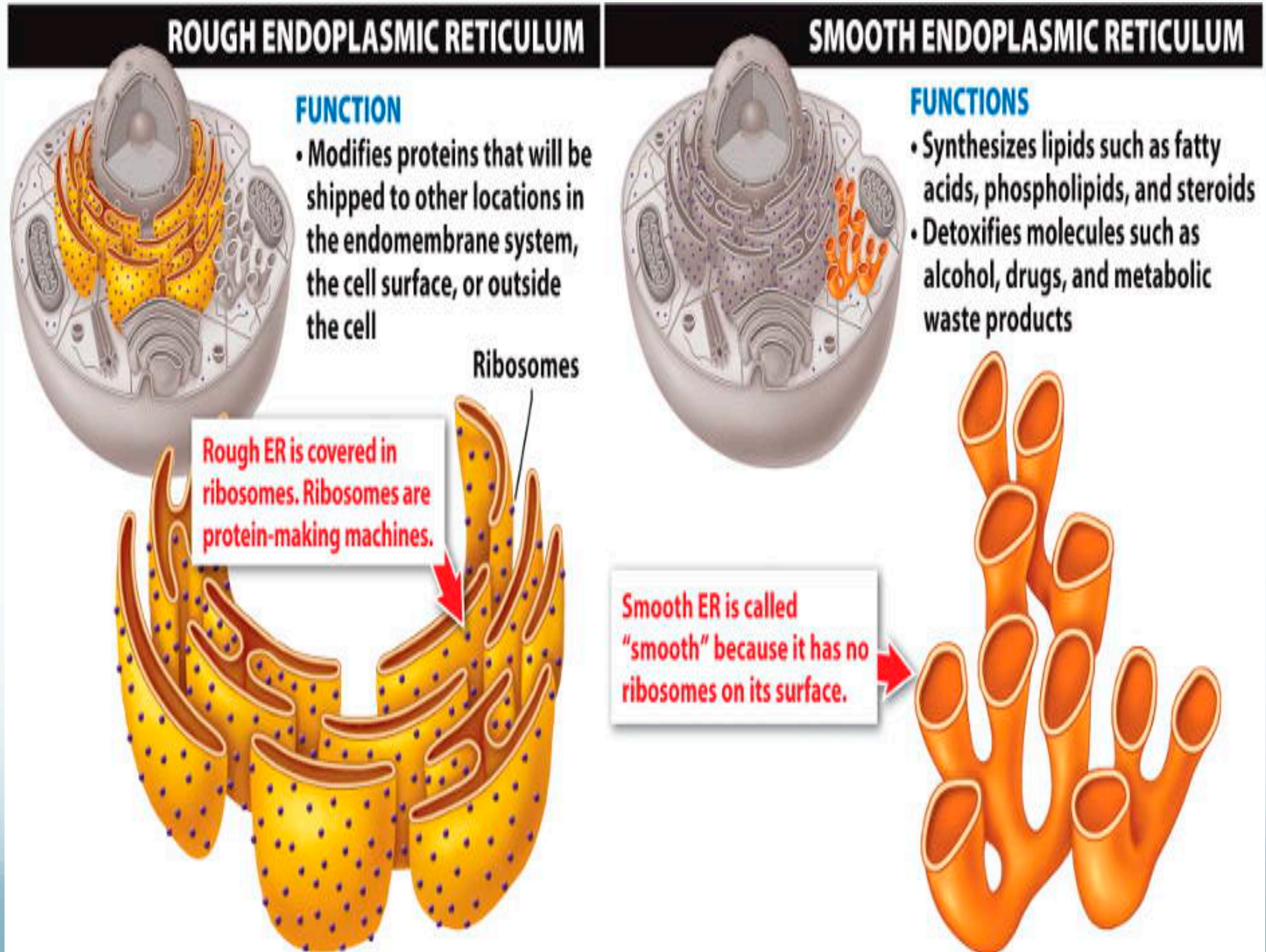
- **Carbohydrate metabolism**
- Glucose is the main source of energy in eukaryotes. Organisms have evolved capabilities to metabolize glucose from other non-carbohydrate precursors in the absence of glucose.
- Liver cells provide one example of the role of smooth endoplasmic reticulum in carbohydrate metabolism, liver cells store carbohydrate In the form of glycogen a polysaccharide. Hydrolysis of glycogen leads to the release of glucose from the liver cells, which is important in the regulation of sugar concentration in the blood.
- The synthesis of glucose from non-carbohydrate precursors like pyruvate, oxaloacetate, succinate, lactate, etc. is known as **gluconeogenesis**.

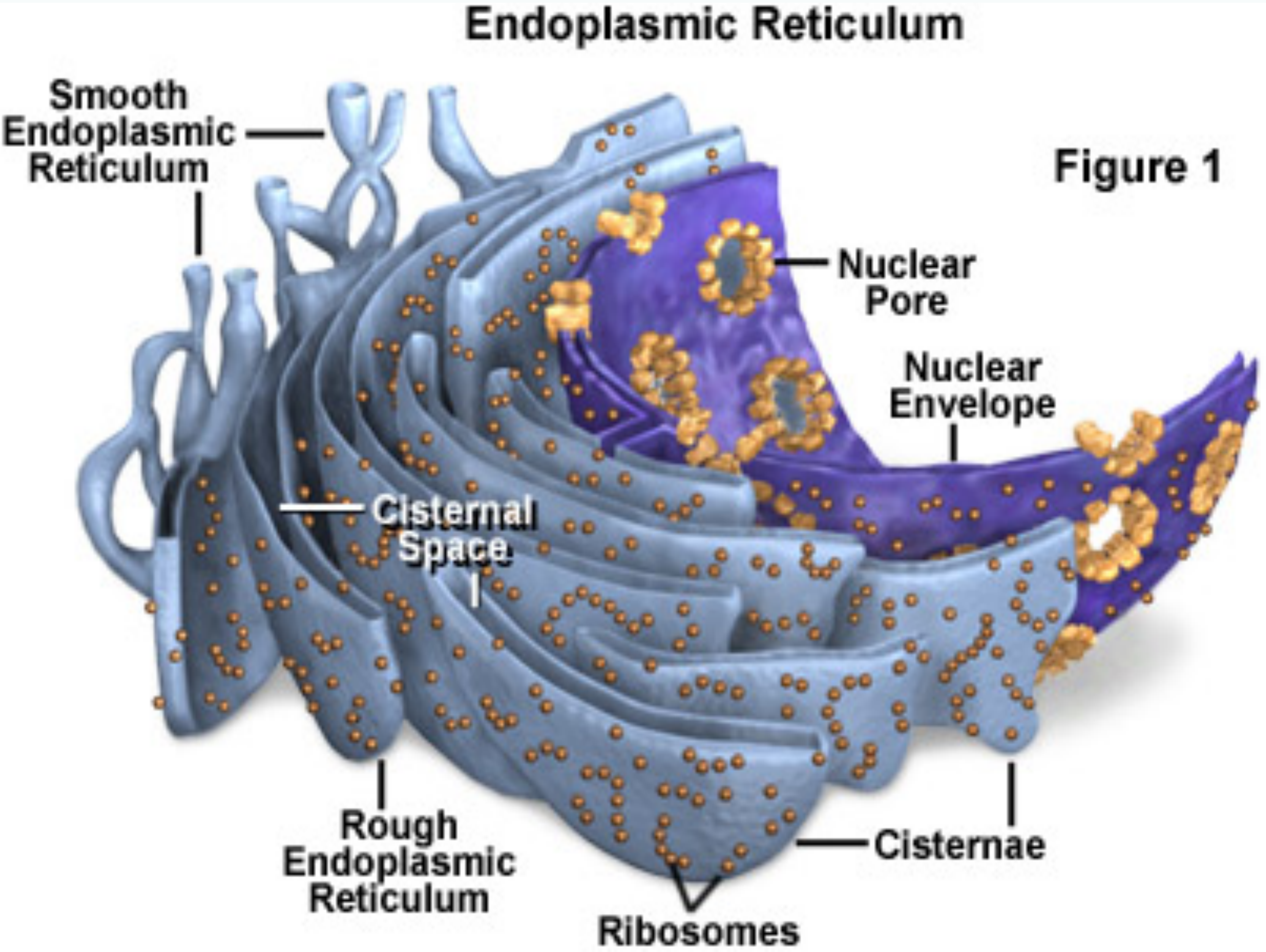
- In most tissues, the final compound formed is glucose 6-phosphate and free glucose is not generated. Glucose 6-phosphate cannot diffuse out of the cell, therefore it is stored inside the cell. A special enzyme called glucose 6-phosphatase which is produced by the smooth endoplasmic reticulum is required to hydrolyze glucose 6-phosphate into glucose.

- This enzyme is only present in tissues of the liver and kidney, where they are involved in the maintenance of blood glucose homeostasis. In these tissues, glucose-6-phosphate is transported to the lumen of the smooth endoplasmic reticulum and finally converted into glucose by the action of the *glucose-6-phosphatase* enzyme.

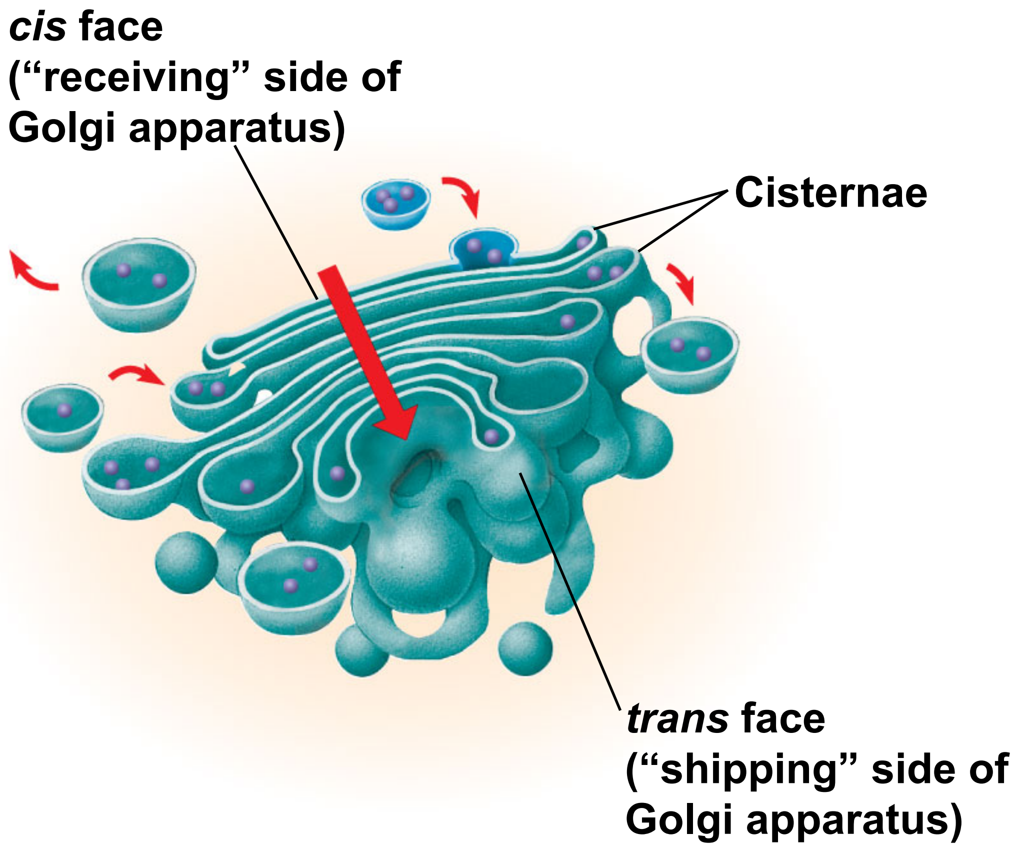
- **Regulation of calcium ion concentrations**
- The smooth endoplasmic reticulum plays a key role in calcium homeostasis. It is responsible for calcium ion storage and release. In muscle cells, for example, a special type of smooth endoplasmic reticulum is referred to as **sarcoplasmic reticulum**. The sarcoplasmic reticulum plays an important intracellular calcium.
- When the muscle cells are stimulated by a nerve impulse, calcium ions rush across the membrane of the endoplasmic reticulum into the cytosol and trigger the contraction of muscle cells.

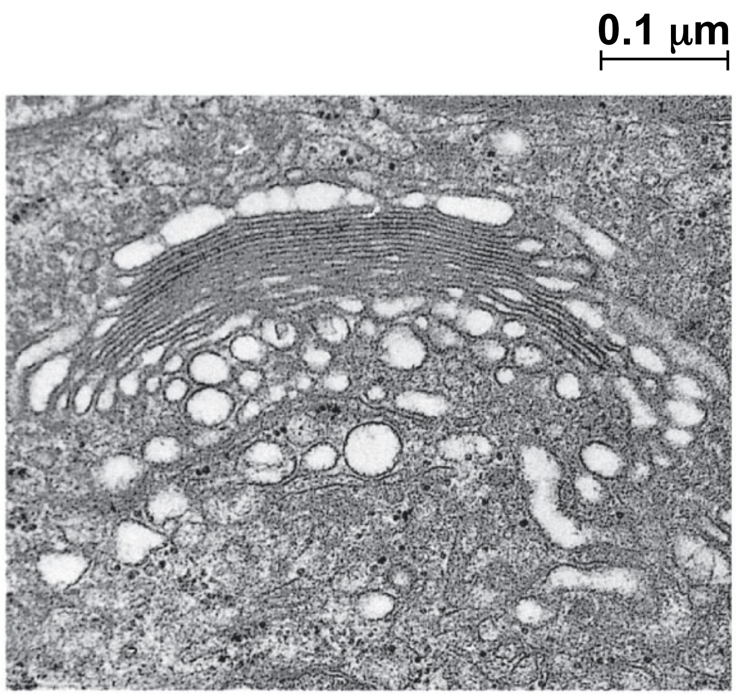
- **Drug detoxification**
- Cytochrome P450s are a large family of enzymes resident in the membrane of the smooth endoplasmic reticulum.
- These enzymes of the smooth endoplasmic reticulum help detoxify drugs and poisons, especially in liver cells.
- Detoxification by smooth ER usually involves adding a hydroxyl group to drug molecules, thus making them more soluble and easier to flush from our bodies.


ROUGH ENDOPLASMIC RETICULUM
FUNCTION
• Modifies proteins that will be shipped to other locations in the endomembrane system, the cell surface, or outside the cell
Ribosomes
Rough ER is covered in ribosomes. Ribosomes are protein-making machines.
SMOOTH ENDOPLASMIC RETICULUM
FUNCTIONS
• Synthesizes lipids such as fatty acids, phospholipids, and steroids
• Detoxifies molecules such as alcohol, drugs, and metabolic waste products
Smooth ER is called "smooth" because it has no ribosomes on its surface.

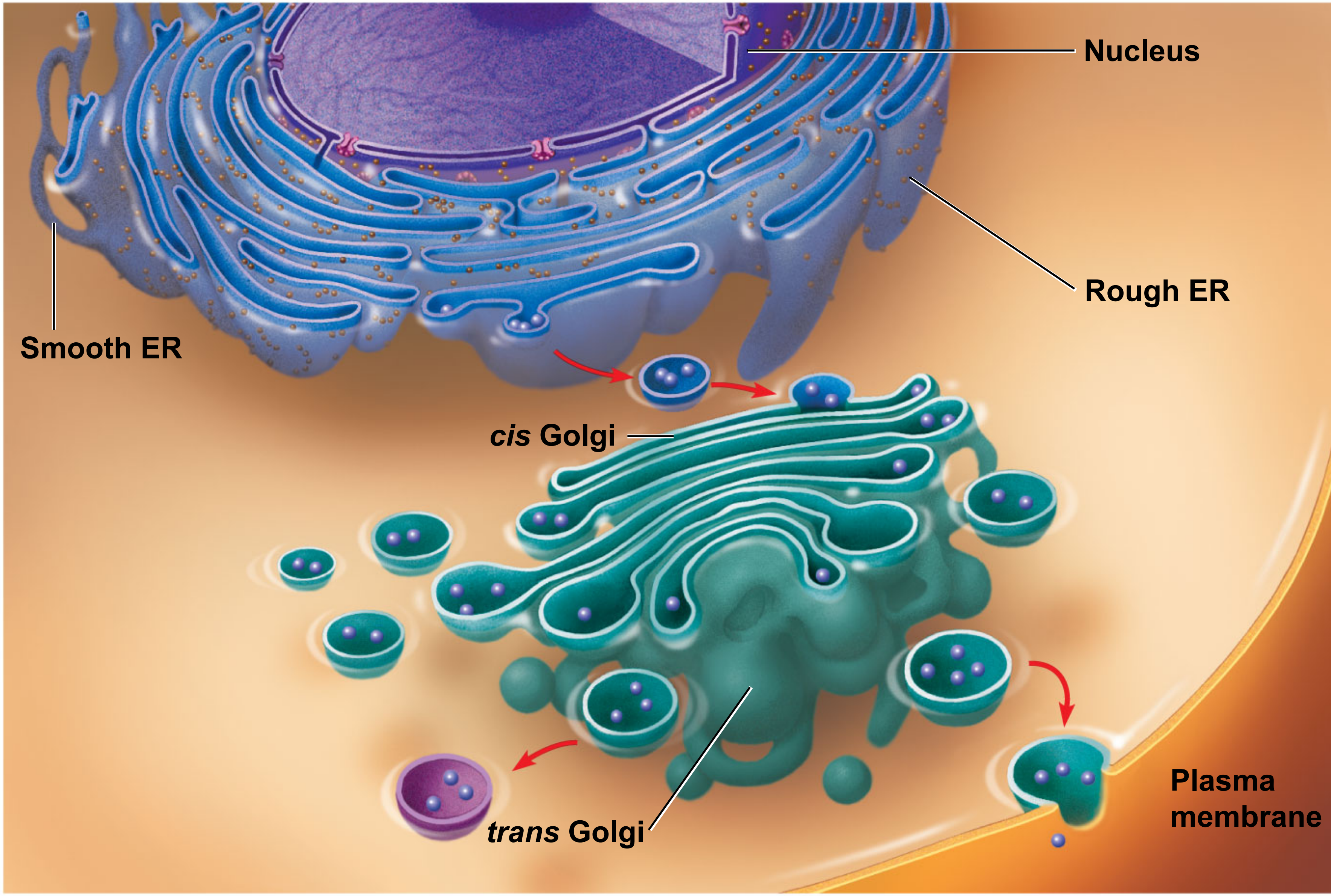

Endoplasmic Reticulum
Figure 1
Smooth Endoplasmic Reticulum
Nuclear Pore
Nuclear Envelope
Cisternal Space
Rough Endoplasmic Reticulum
Cisternae
Ribosomes

- **Golgi apparatus**,
- membrane-bound organelle of eukaryotic cells (cells with clearly defined nuclei) that is made up of a series of flattened, stacked pouches called cisternae.
-
- It is located in the cytoplasm next to the endoplasmic reticulum and near the cell nucleus.
- While many types of cells contain only one or several Golgi apparatus.
- Secretory proteins and glycoproteins, cell membrane proteins, lysosomal proteins, and some glycolipids all pass through the Golgi apparatus at some point in their maturation.





TEM of Golgi apparatus


Nucleus
Rough ER
Smooth ER
cis Golgi
trans Golgi
Plasma
membrane

- The Golgi apparatus itself is structurally polarized, with three primary compartments lying between the "cis" face and the "trans" face.

- The three primary compartments of the apparatus are known generally as "cis" (cisternae nearest the endoplasmic reticulum), "medial" (central layers of cisternae), and "trans" (cisternae farthest from the endoplasmic reticulum).

- The proteins and lipids received at the cis face arrive in clusters of fused vesicles.

- These fused vesicles migrate along microtubules through a special trafficking compartment, called the vesicular-tubular cluster, that lies between the endoplasmic reticulum and the Golgi apparatus

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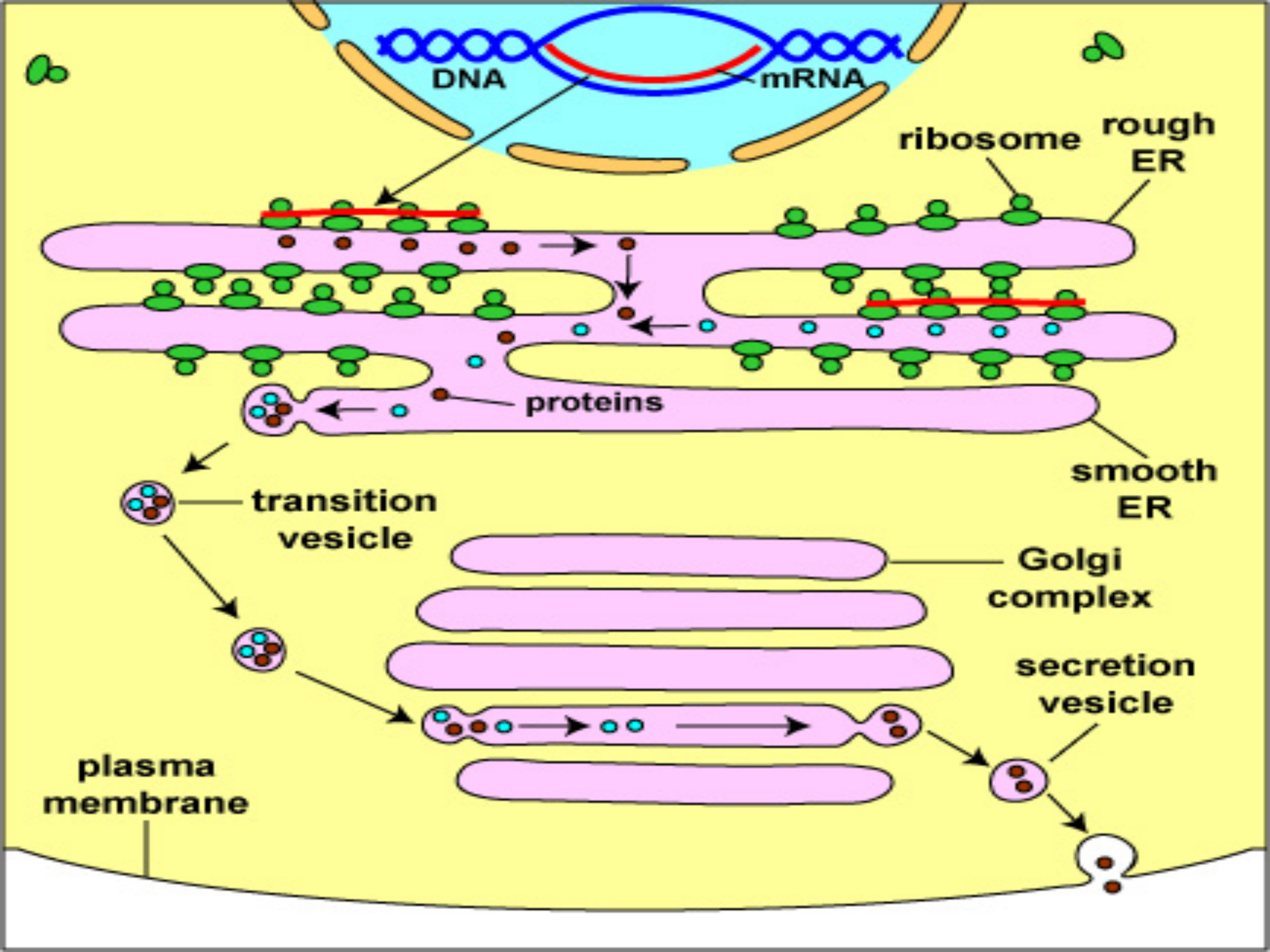
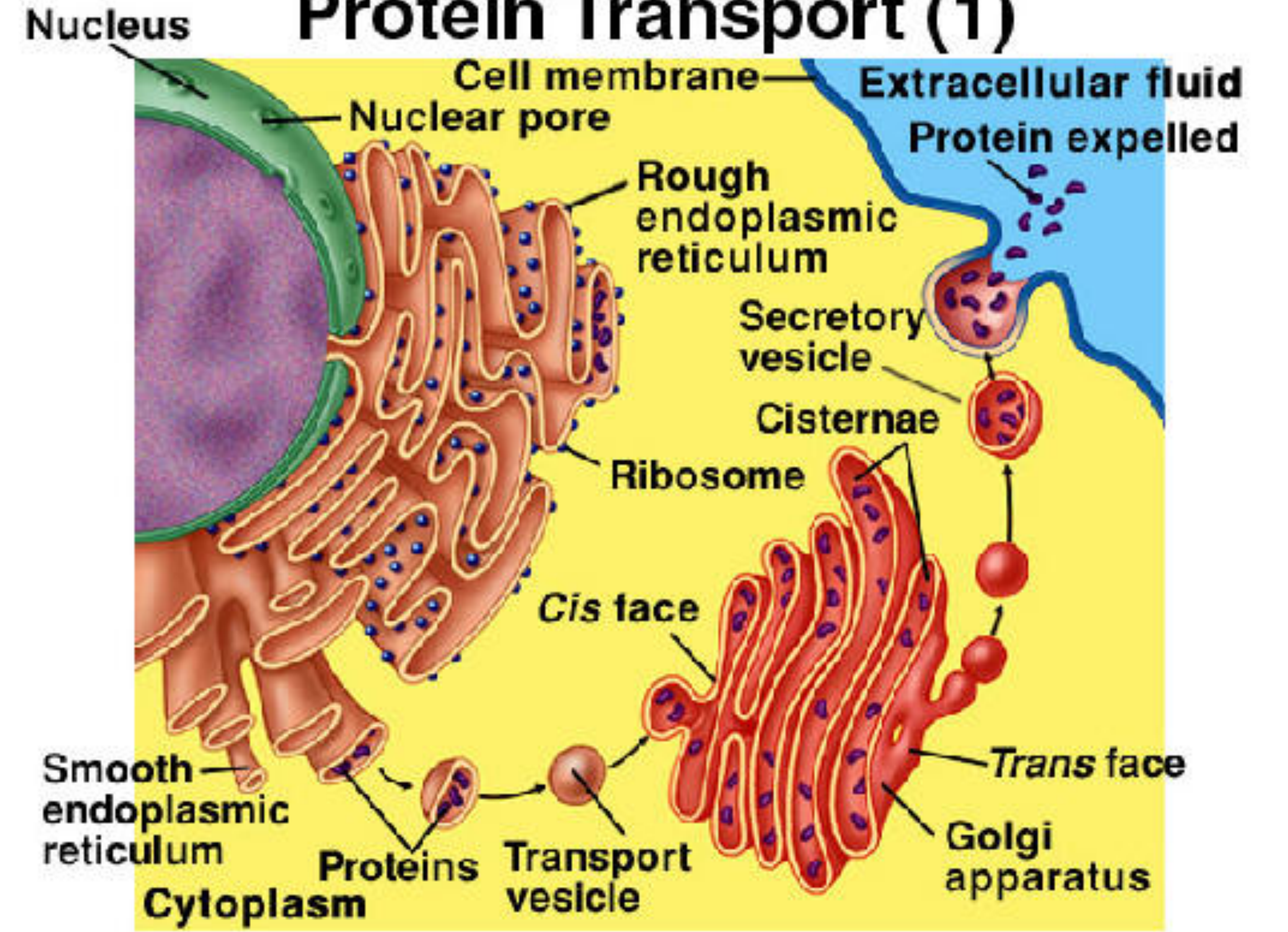
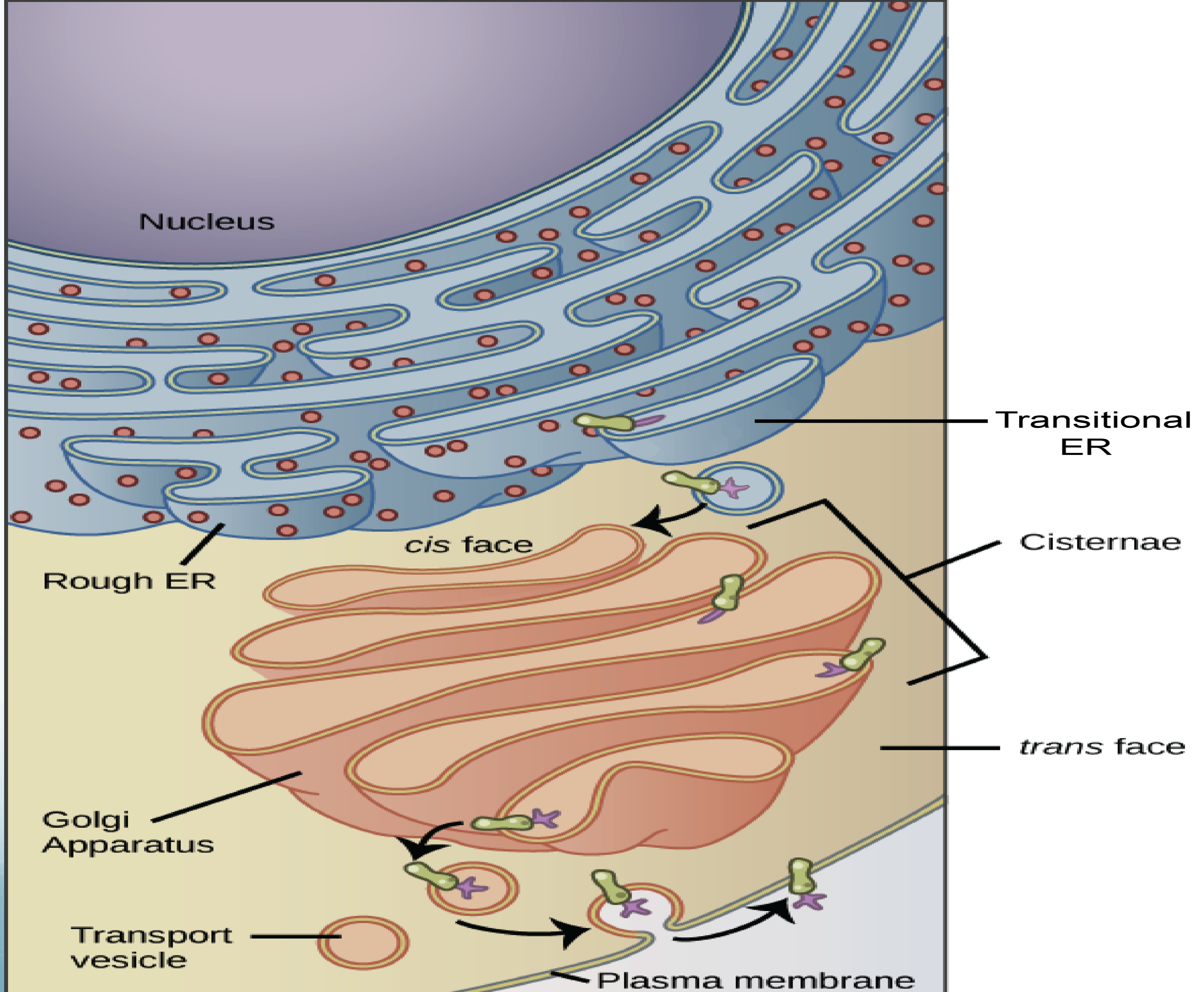
- When a vesicle cluster fuses with the cis membrane, the contents are delivered into the lumen of the cis face cisterna.
- As proteins and lipids progress from the cis face to the trans face, they are modified into functional molecules and are marked for delivery to specific intracellular or extracellular locations.
- **Some modifications involve cleavage of oligosaccharide side chains followed by attachment of different sugar in place of the side chain.**
- **Other modifications may involve the addition of fatty acids or phosphate groups (phosphorylation) or the removal of monosaccharaides.**

- The different enzyme-driven modification reactions are specific to the compartments of the Golgi apparatus.
- In the final stage of transport through the Golgi apparatus, **modified proteins and lipids are sorted in the trans Golgi network and are packaged into vesicles at the trans face.**
- These vesicles then deliver the molecules to their target destinations, such as **lysosomes or the cell membrane.**

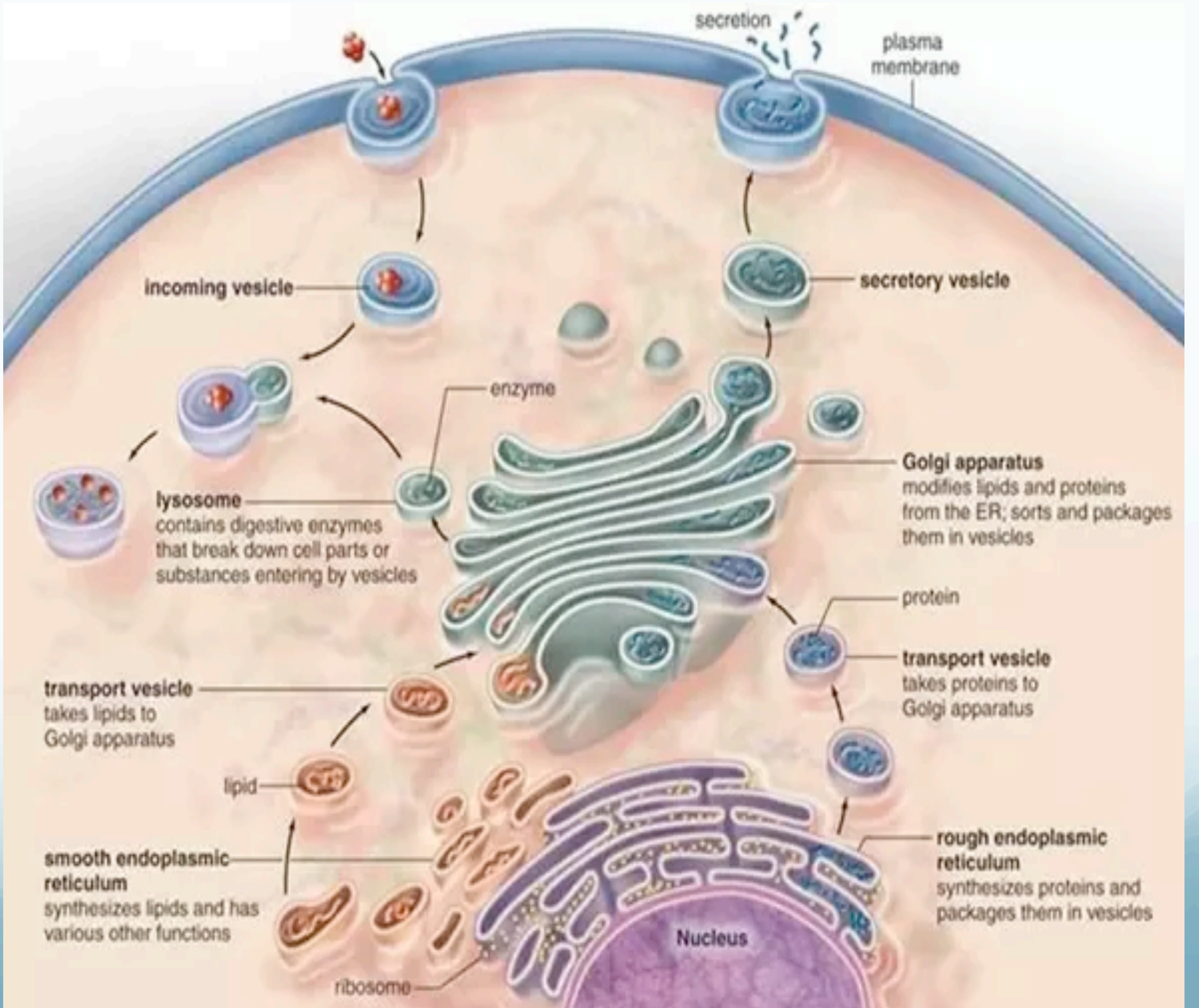

DNA
mRNA
ribosome
rough ER
proteins
smooth ER
transition vesicle
Golgi complex
secretion vesicle
plasma membrane

Protein Transport (1)

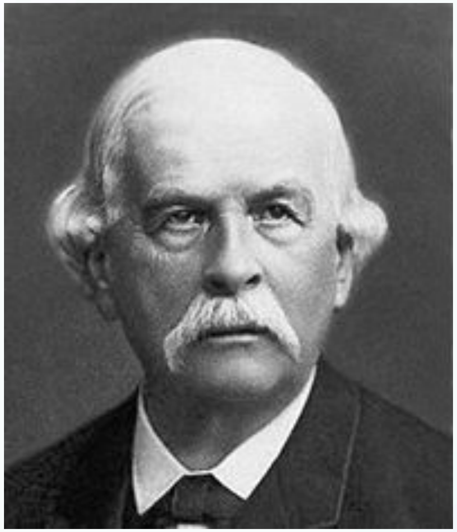




Nucleus
Transitional ER
cis face
Rough ER
Cisternae
trans face
Golgi Apparatus
Transport vesicle
Plasma membrane

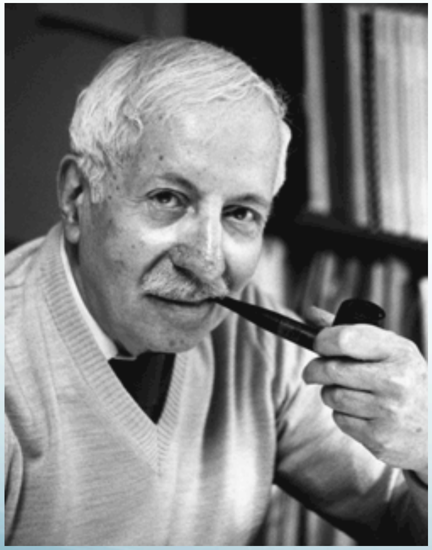

secretion
plasma membrane
incoming vesicle
secretory vesicle
enzyme
Golgi apparatus
modifies lipids and proteins from the ER; sorts and packages them in vesicles
lysosome
contains digestive enzymes that break down cell parts or substances entering by vesicles
protein
transport vesicle
takes proteins to Golgi apparatus
transport vesicle
takes lipids to Golgi apparatus
lipid
smooth endoplasmic reticulum
synthesizes lipids and has various other functions
rough endoplasmic reticulum
synthesizes proteins and packages them in vesicles
Nucleus
ribosome

Mitochondrion



established them as cell organelles
Richard Altmann (1886) german pathologist

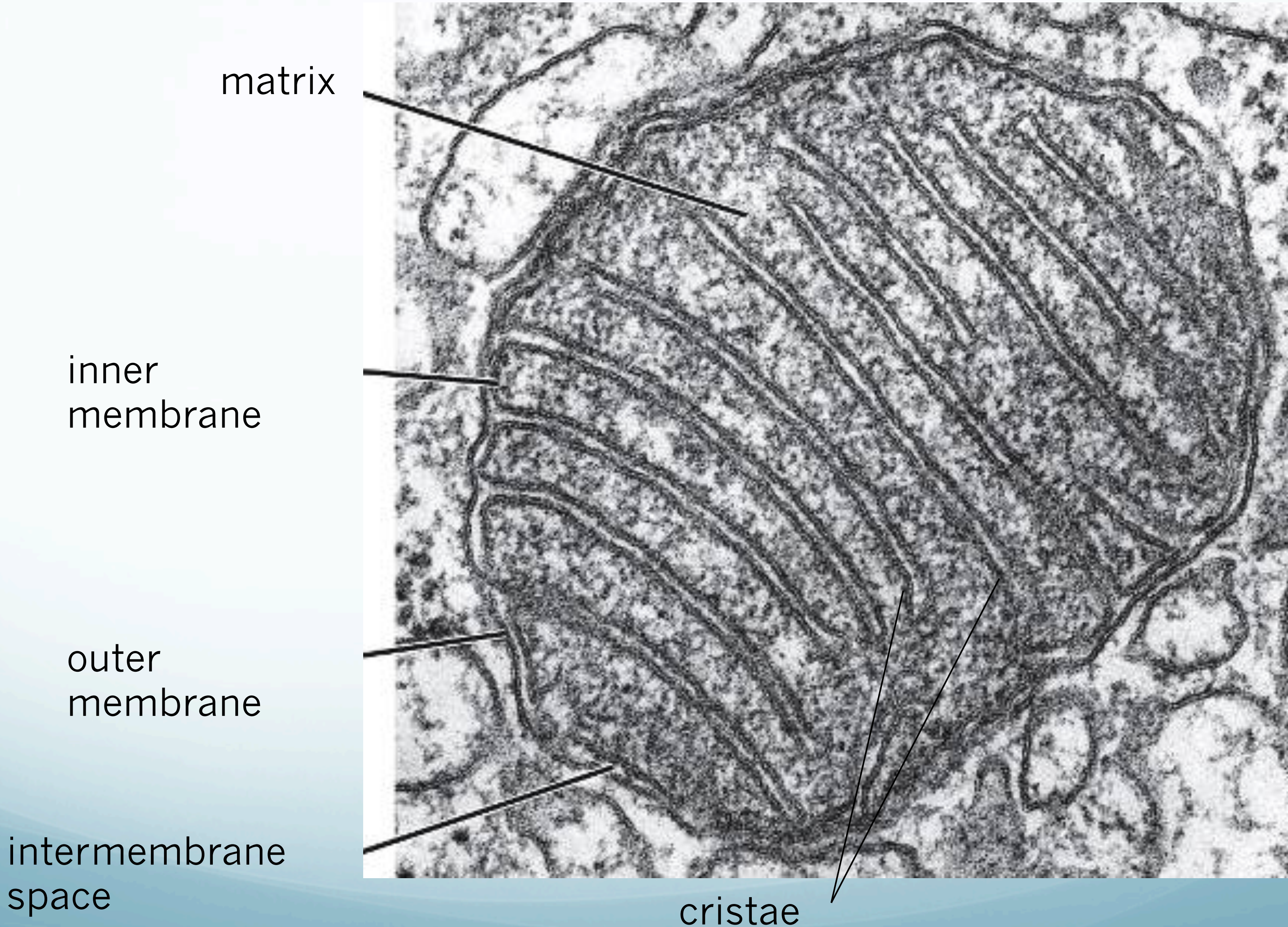
mitos= "thread”   *chondrion*= "granule"



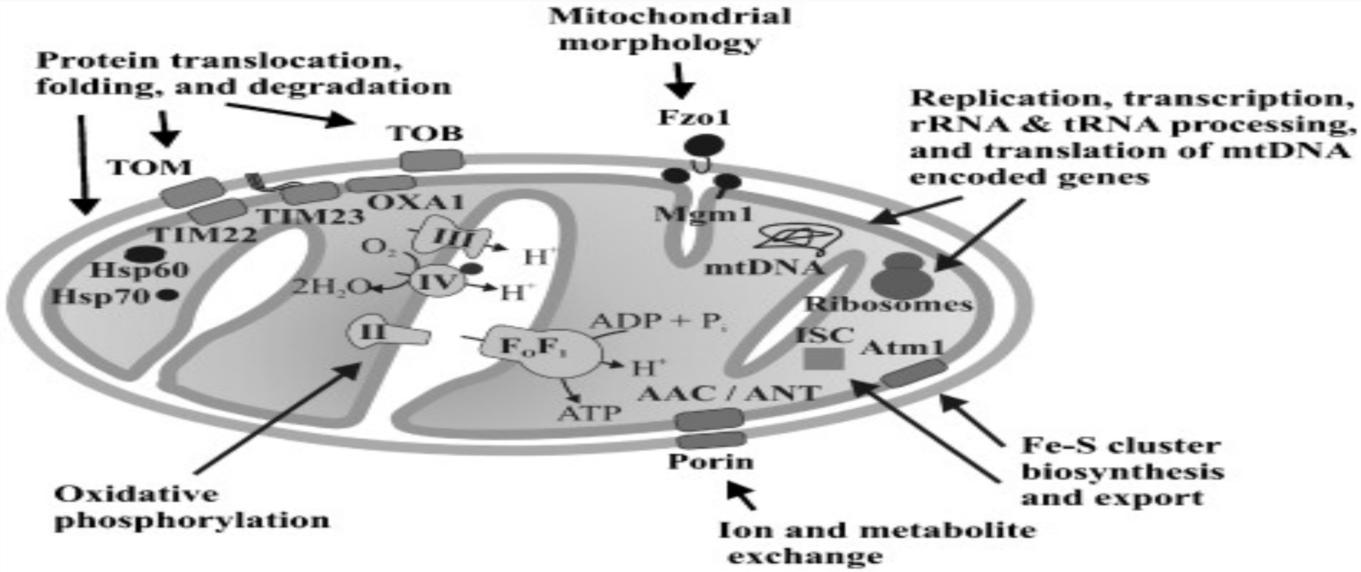
”Powerhouse of the cell”

Philip Siekevitz (1957) american biochemist

Ultrastructure of the Mitochondria

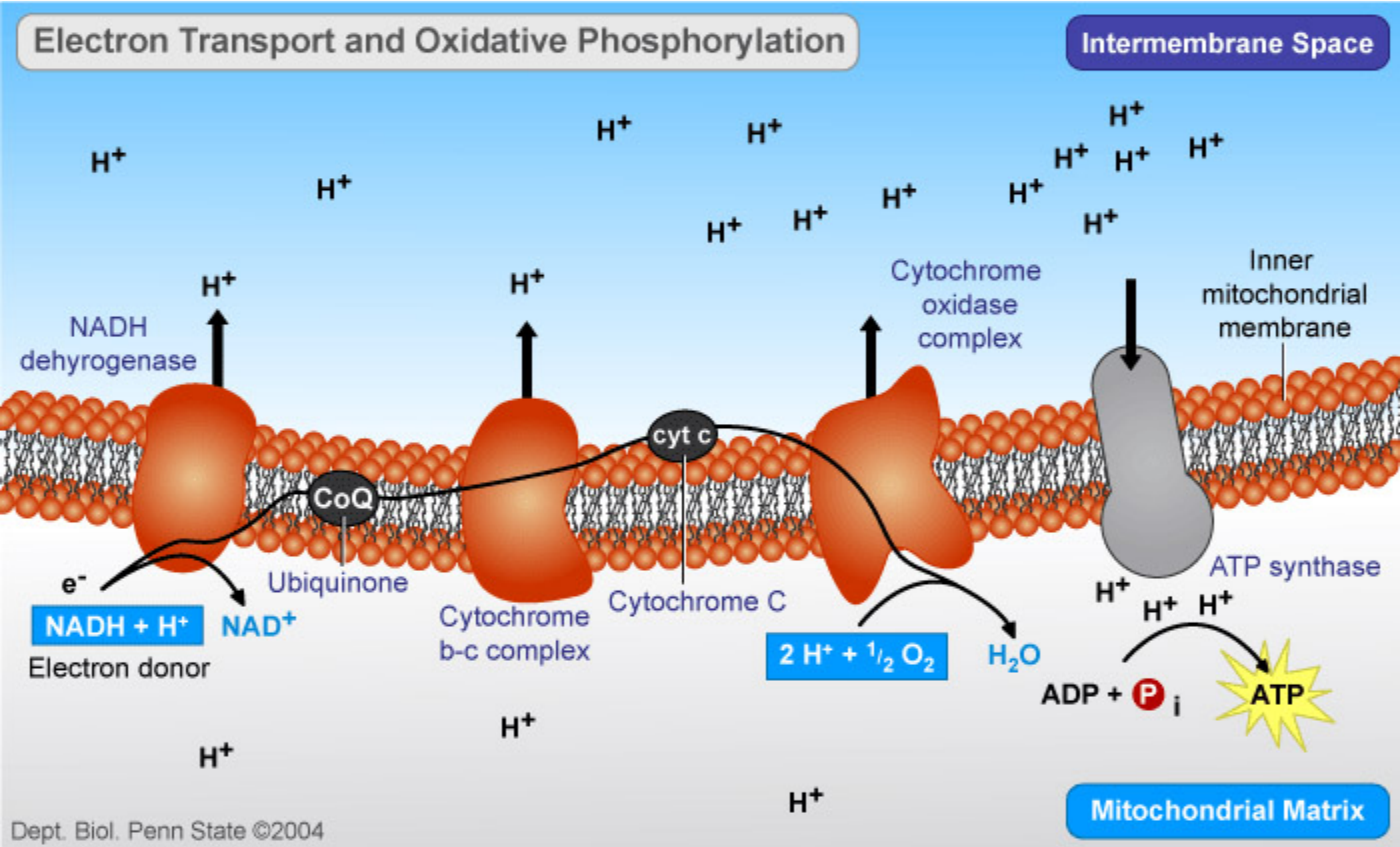


Membranes of the Mitochondria



Outer membrane: smooth, special proteins (porins), translocon complexes and phospholipids. **Permeable to nutrient molecules, ions, ATP and ADP**

Inner membrane: folded (cristae-increased surface), cardiolipin (regulation of membrane permeability of ions) complexes for electron transport, ATP synthase complexes, transfer regulating proteins (TIMs). **Permeable for oxigene, carbon dioxide, water**.



Electron Transport and Oxidative Phosphorylation
Intermembrane Space
H^+
NADH dehyrogenase
Cytochrome oxidase complex
Inner mitochondrial membrane
cyt c
CoQ
e^-
$NADH + H^+$
NAD^+
Electron donor
Ubiquinone
Cytochrome b-c complex
Cytochrome C
$2 H^+ + \frac{1}{2} O_2$
H_2O
ATP synthase
ADP + P i
ATP
Mitochondrial Matrix
Dept. Biol. Penn State ©2004

Major function of the Mitochondria:

-Outer membrane:

-Transport functions, most substances are also transported through the inner membrane.

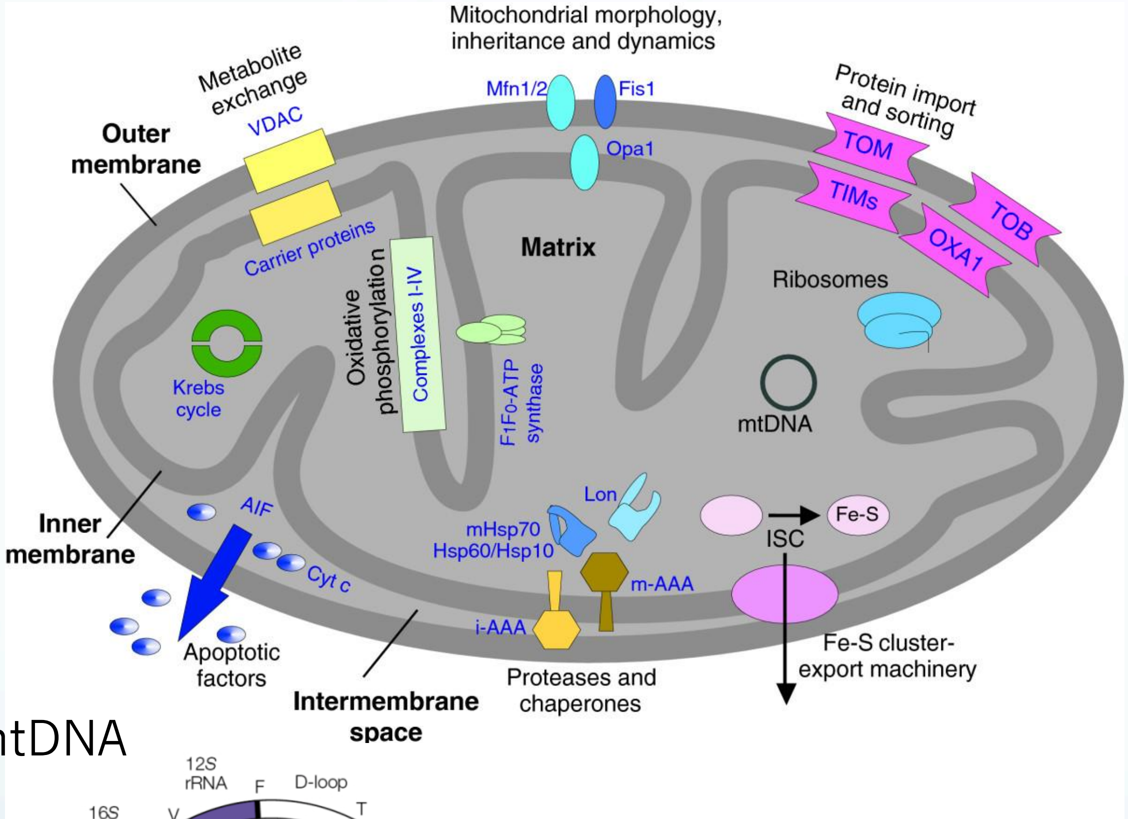
-Inner membrane :

1-Proton-transport and oxidative phosphorylation
2- Ca”+ ,transport (to sequester the Ca^{2+} from the cytoplasm)
3- Lipid, aminoacid, ADP, phosphate, pyruvate transport inward, ATP outward
4- Transports of proteins (mitochondrial enzymes), tRNA (through translocons).

- **-Matrix:**
  1- Citrate cycle (Krebs- or Szentgyörgyi-Krebs-cycle);
  2- Lipid- and aminoacid-degradation to acetyl-groups for the citrate cycle.
  3- Protein- and lipid synthesis (their special lipid is the cardiolipin).
  4- Ca”+ storage.
  Beside these, mitochondria can grow, divide, move (along microtubules), and if their content is released into the cytoplasm, it switches the apoptotic cascade

Other functions: heat production, involvement in steroid synthesis and apoptosis

Mitochondrial matrix



Enzymes

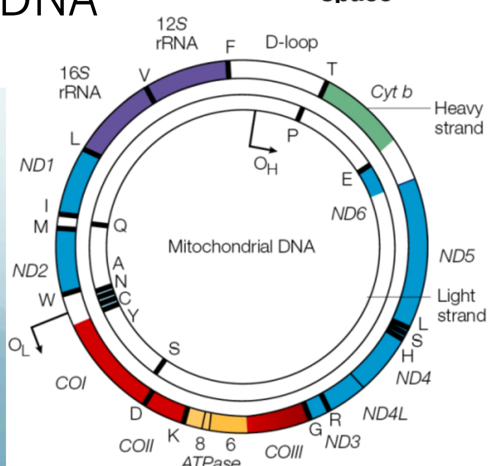
Special mitochondrial ribosomes

tRNAs

DNA

Carbon dioxide and oxygen

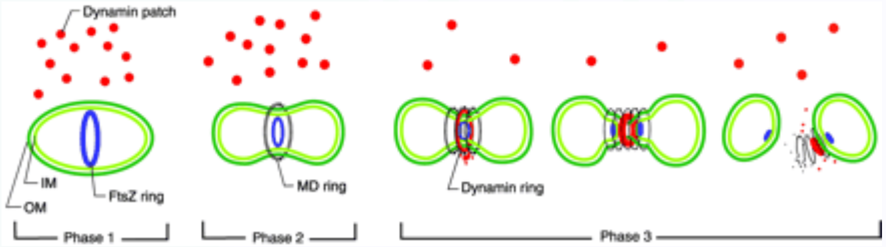
mtDNA

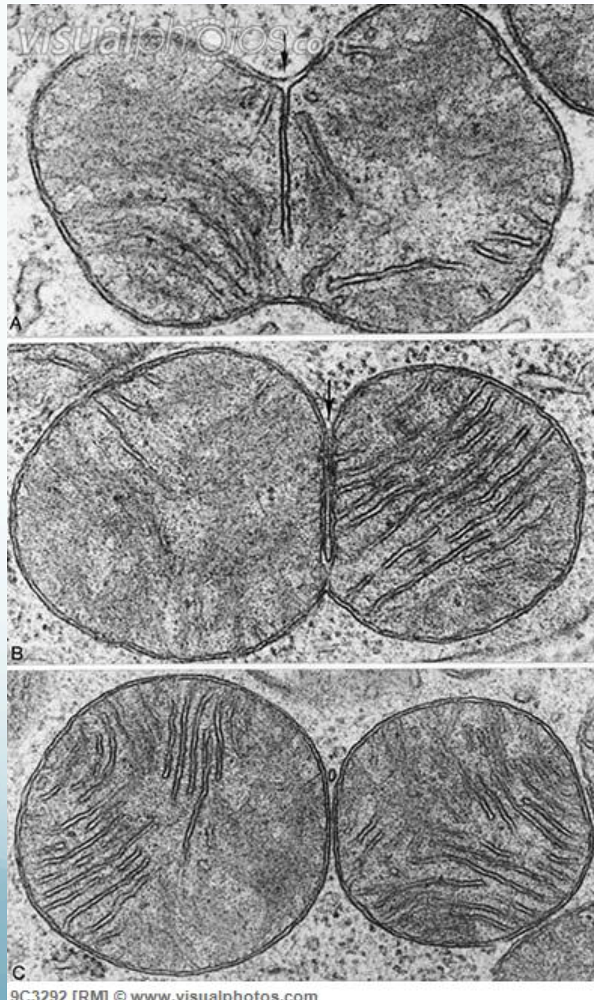


37 genes → 13 genes enzymes for oxidative phosphorilation, others for tRNA and rRNA

The majority of proteins in mitochondria are coded by nuclear DNA

Mitochondrial division

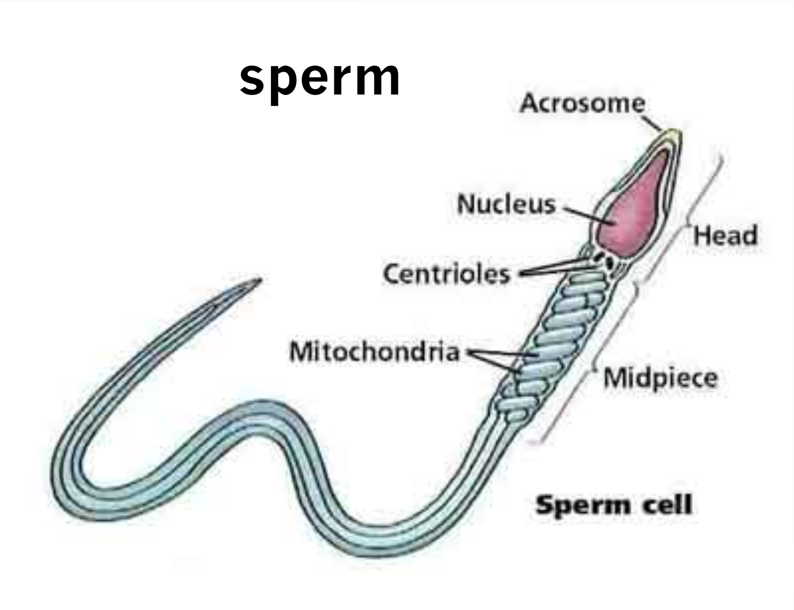


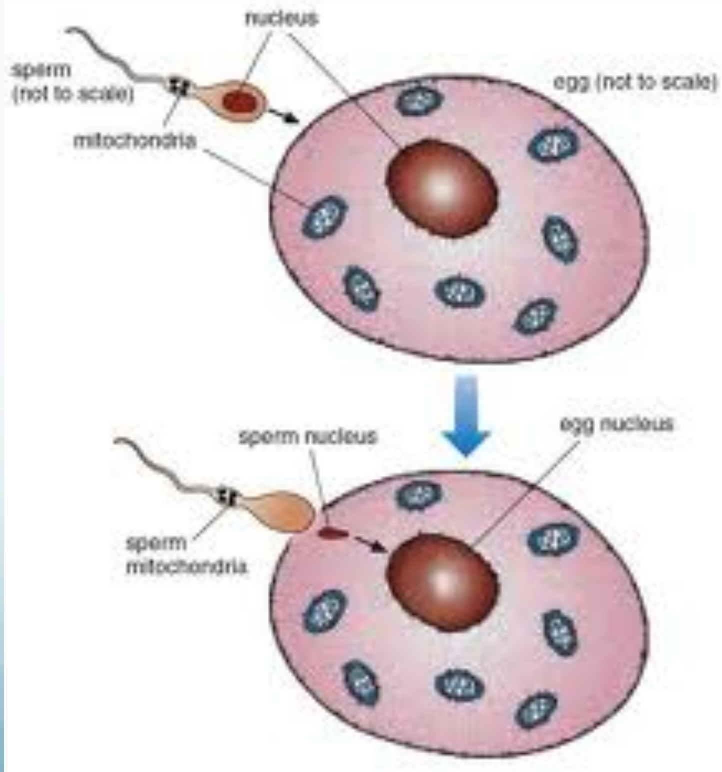


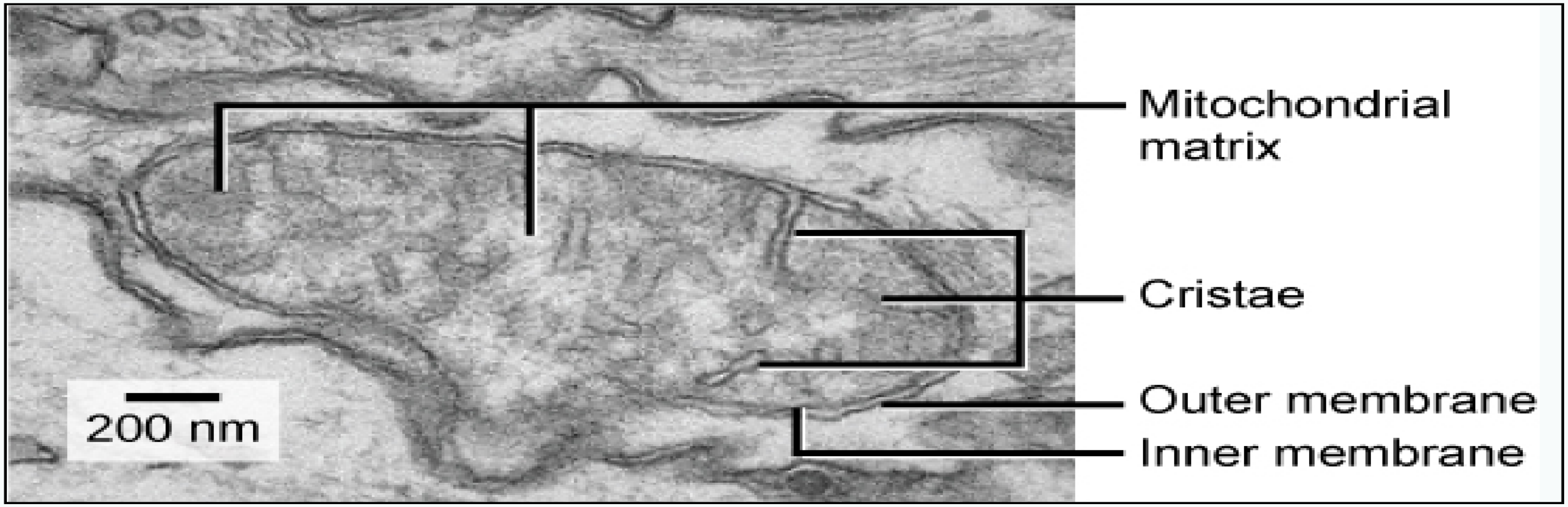
No de novo formation of mitochondria

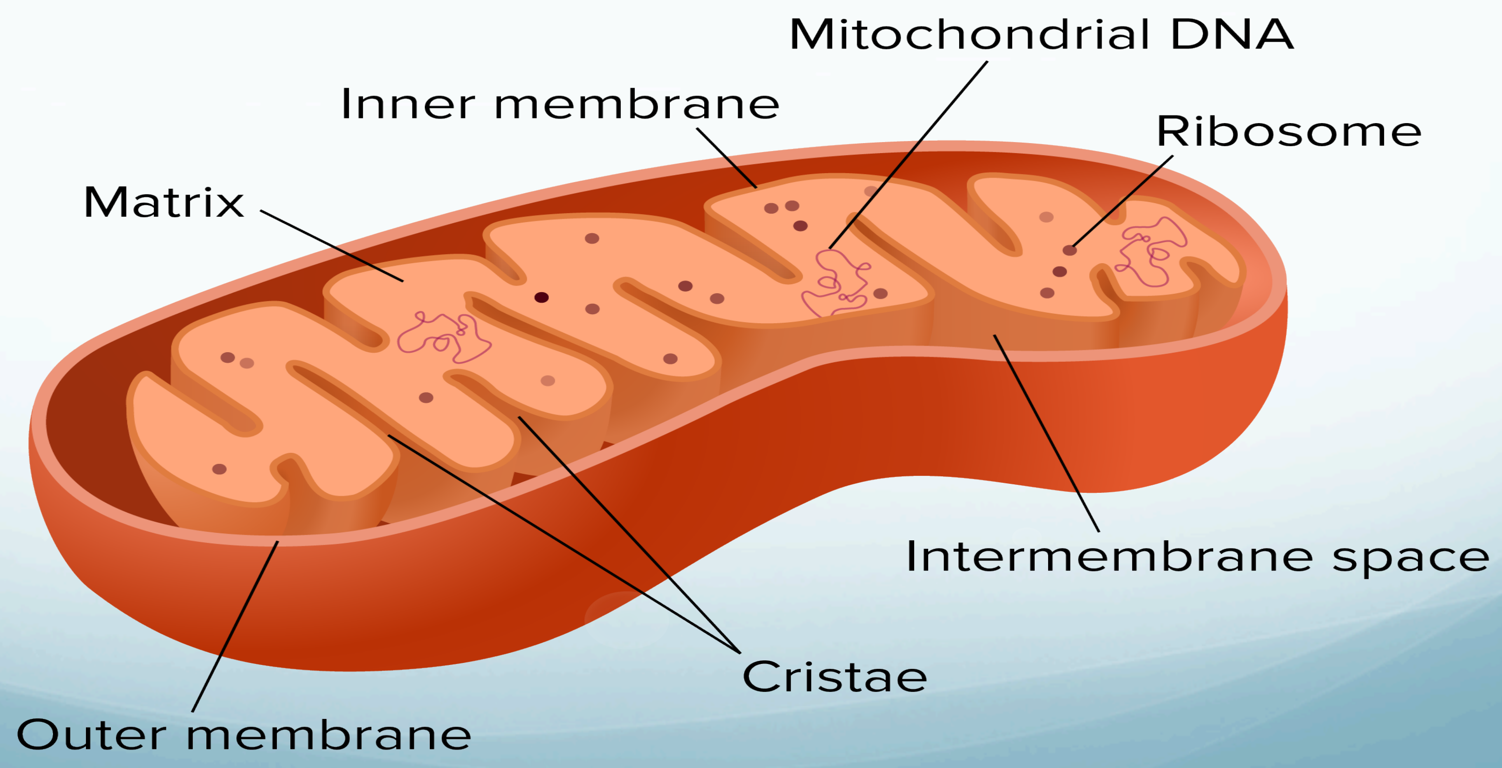
1. Replication of genom
2. Growth
3. Fission
4. Segregation

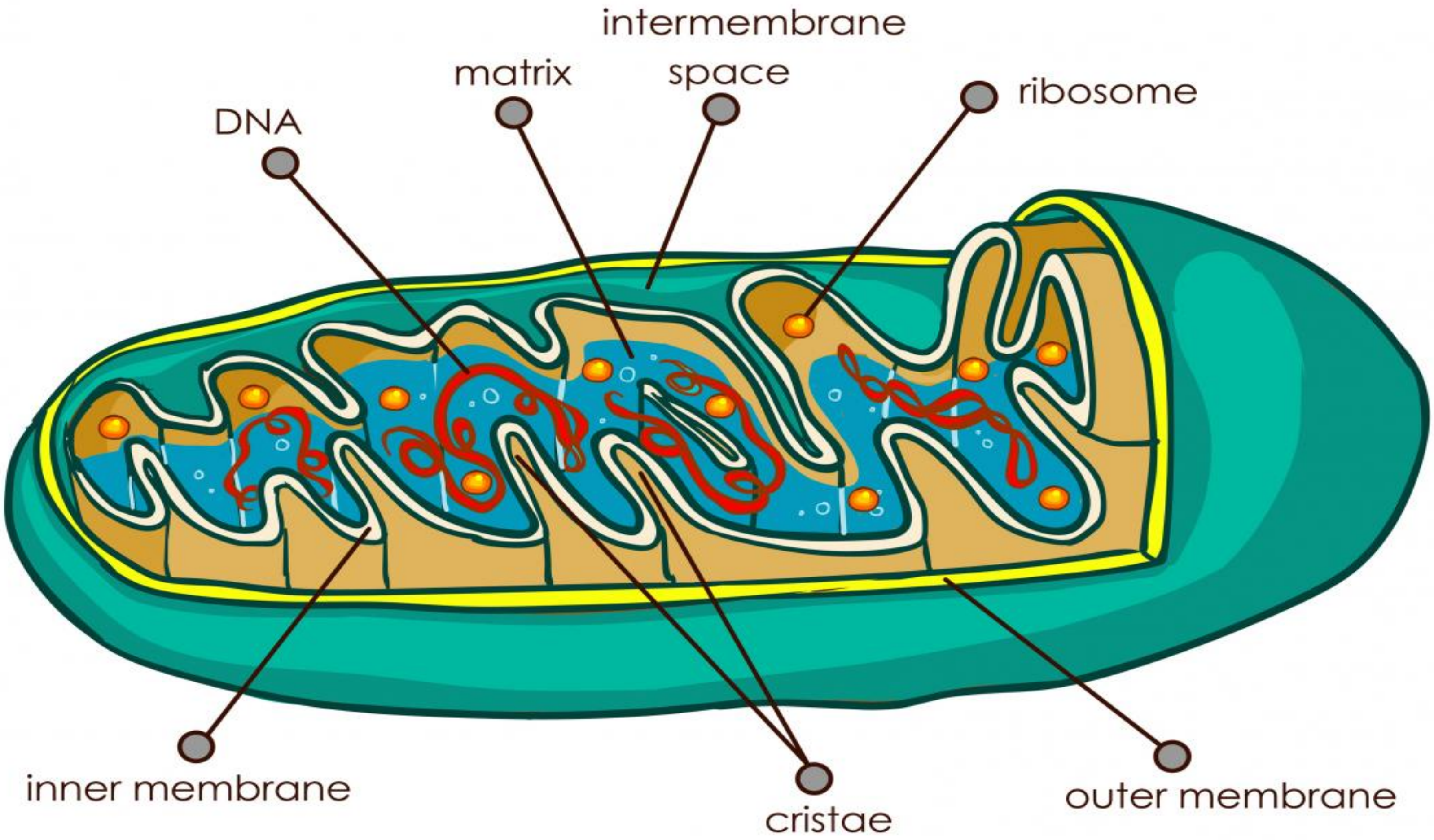
Mitochondria are maternally inherited

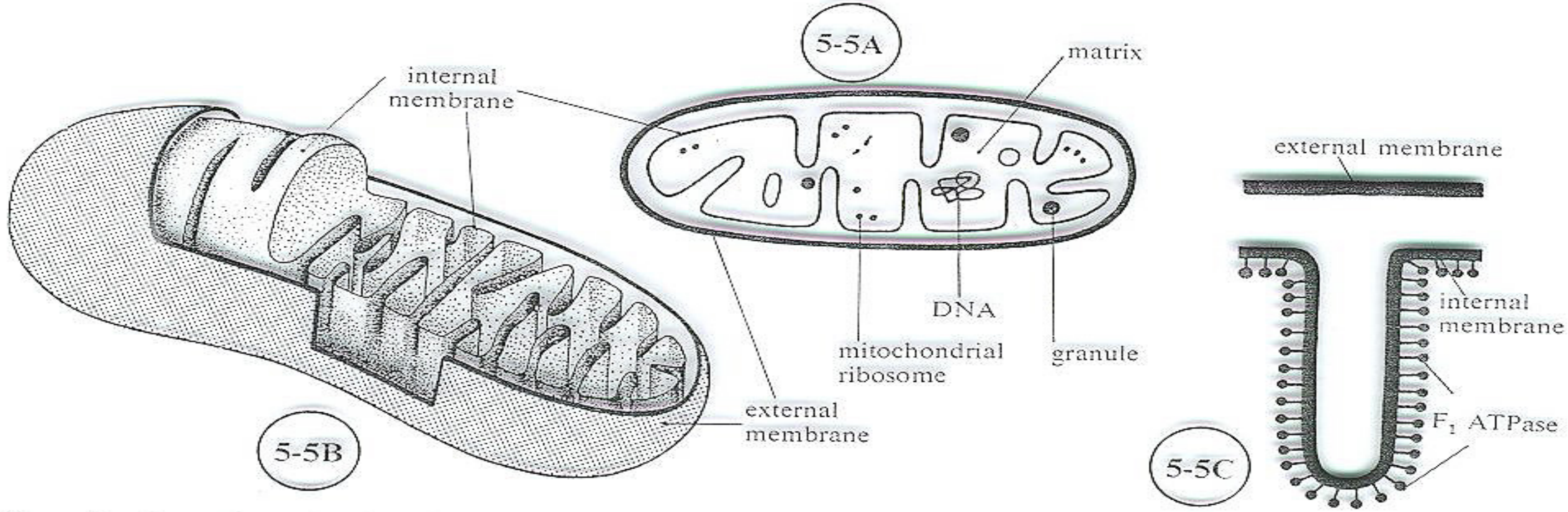




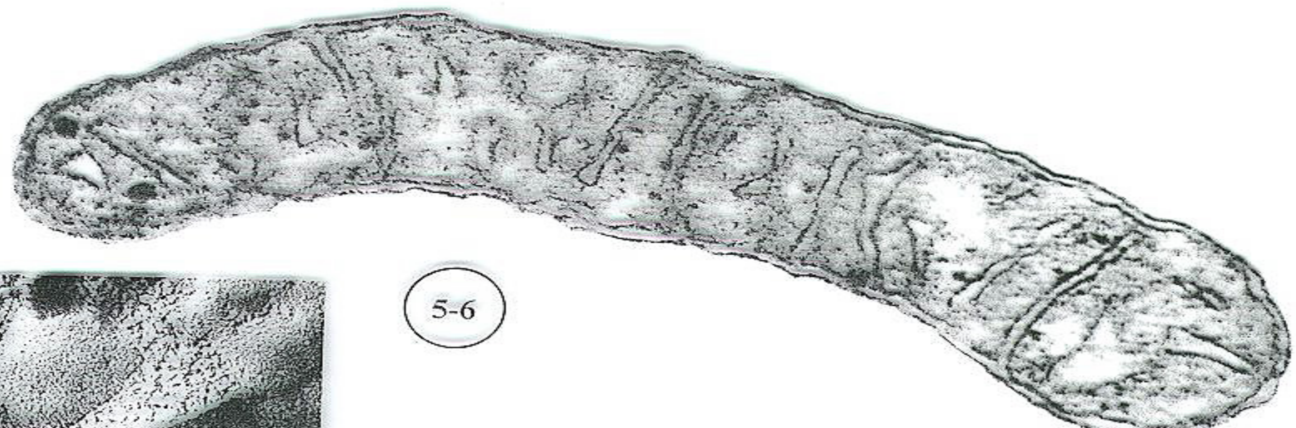

Mitochondrial matrix
Cristae
Outer membrane
Inner membrane
200 nm


Mitochondrial DNA
Inner membrane
Ribosome
Matrix
Intermembrane space
Cristae
Outer membrane


intermembrane
space
matrix
DNA
ribosome
inner membrane
cristae
outer membrane



5-5. Organization of a mitochondrion.
(A) Section. (B) Schematic three-dimensional representation. (C) Detail of a crista, showing the pedunculated particles (ATPosomes) of the internal membrane.



5-6. Mitochondrion in an ultrathin section.

5-7. Mitochondrion after freeze-etching. $\times$30,000. The fracture shows the organelle both from the surface and in section.

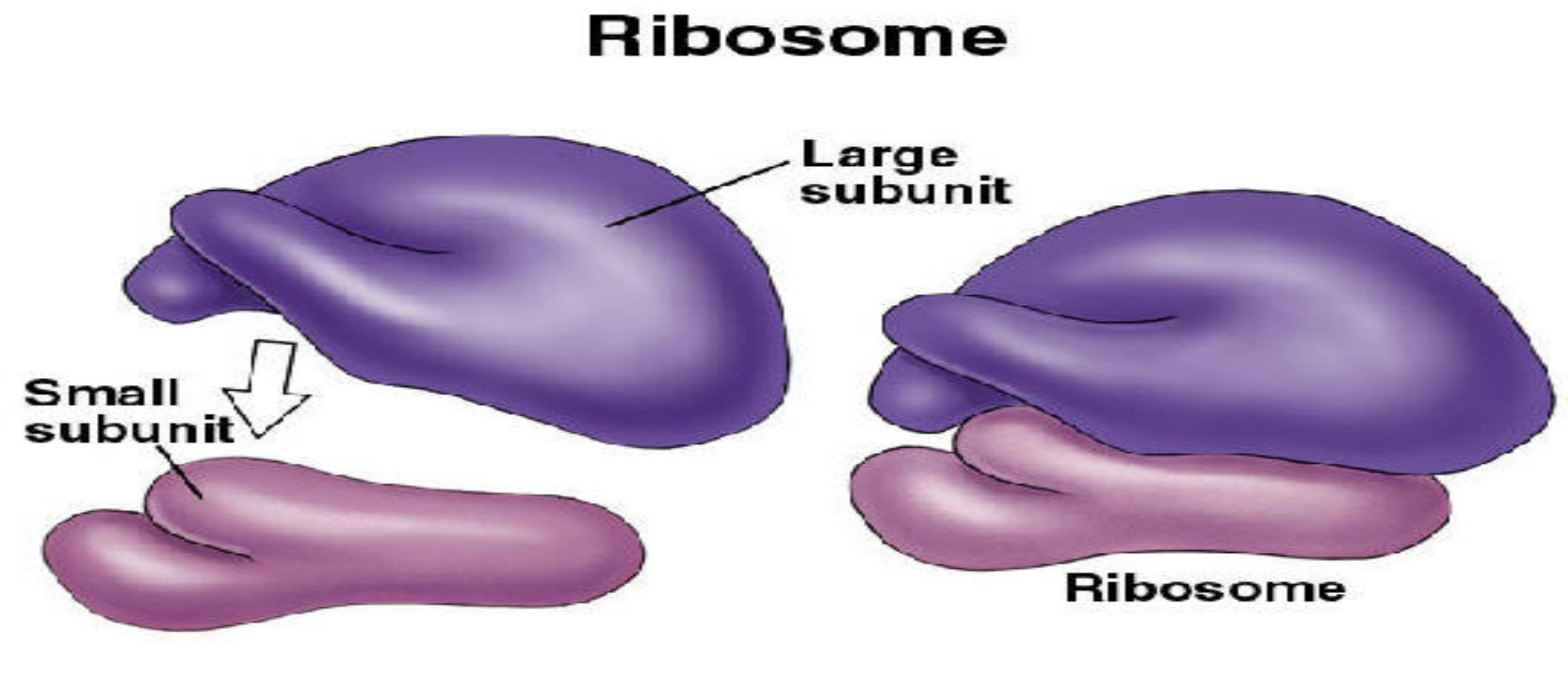
4-Ribosomes

Structure - cells normally have millions of ribosomes, each ribosome has two parts which come together during protein synthesis.

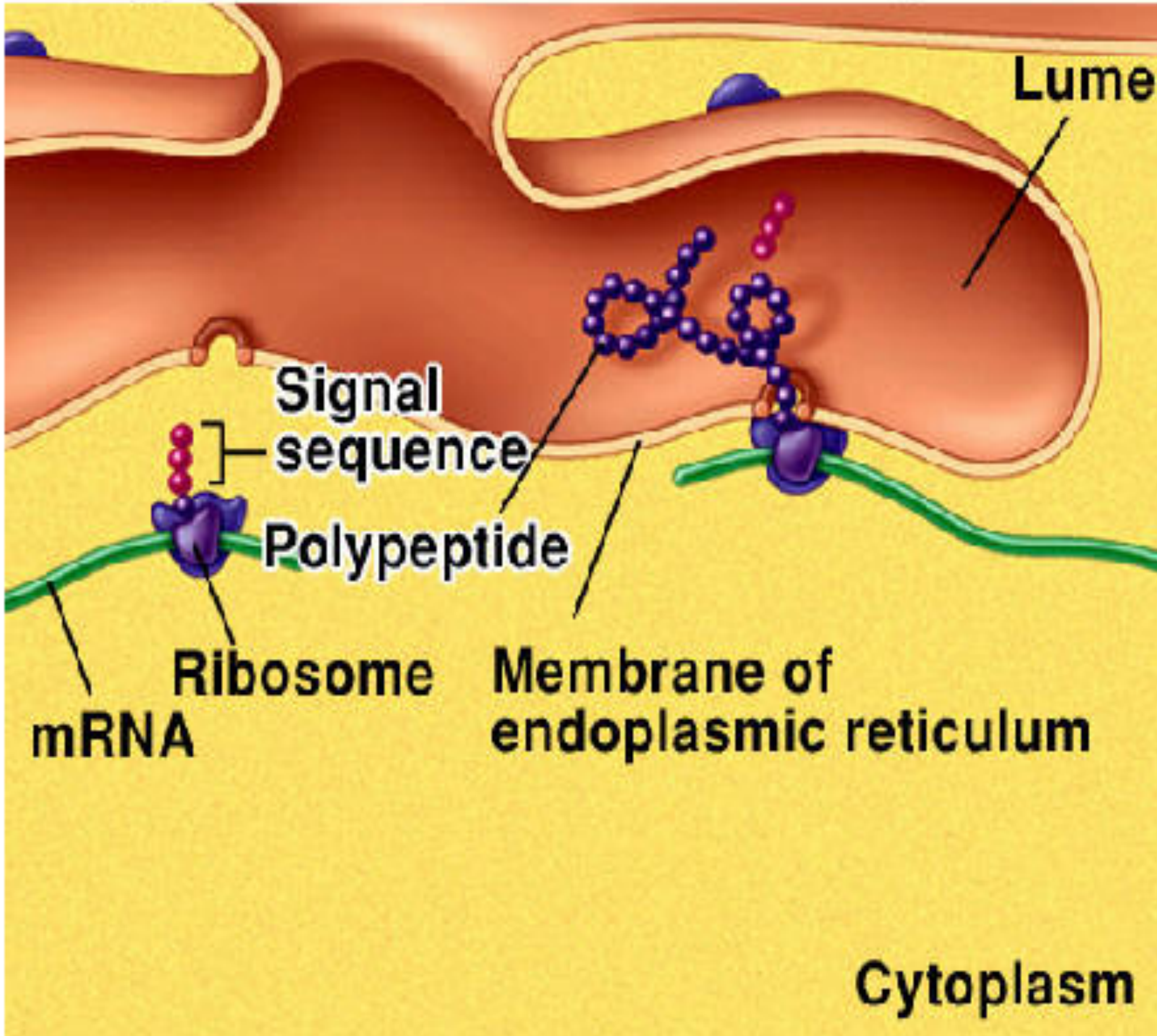
- A ribosome is made of numerous proteins and RNA of approximately 60 percent RNA and 40 percent protein. In eukaryotes, ribosomes are made of four strands of RNA. In prokaryotes, they consist of three strands of RNA only.

Ribosomes

Function- ribosomes are responsible for protein synthesis

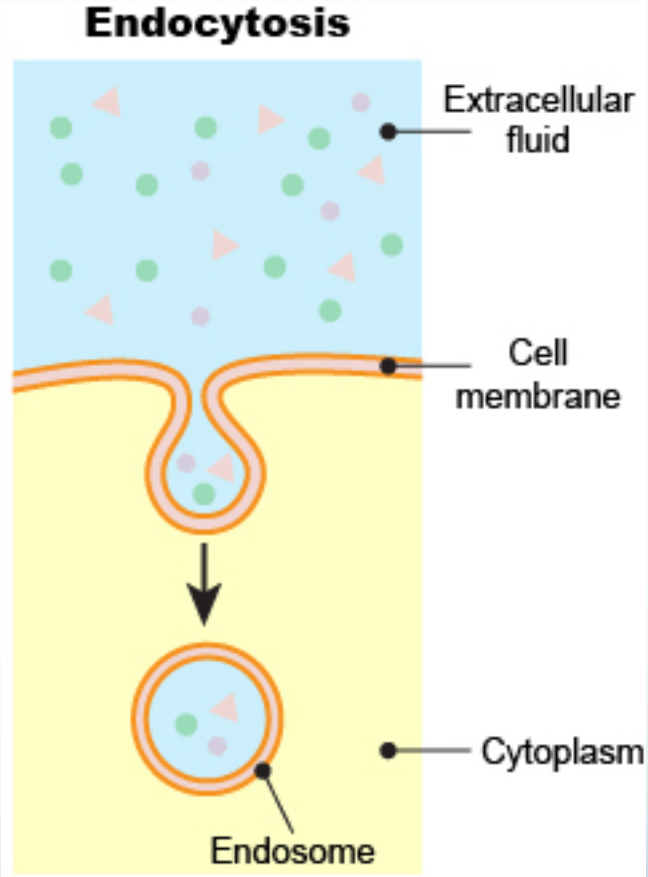


Rough ER and Protein Synthesis



5-Endosomes

- Endosomes are membrane-bound vesicles, formed via a complex family of processes collectively known as endocytosis, and found in the cytoplasm of virtually every animal cell. The basic mechanism of endocytosis is the reverse of what occurs during exocytosis or cellular secretion. It involves the invagination (folding inward) of a cell's plasma membrane to surround macromolecules or other matter diffusing through the extracellular fluid.


Endocytosis
Extracellular fluid
Cell membrane
Cytoplasm
Endosome


THANK YOU!

- What is cytoplasm?
- What is endomembrane system?
- What is Endoplasmic reticulum(ER), from what its mainly composed?
- What are different types of ER?
- Differentiate between different types of ER :
- A- according to structure
- B- according to function
- C- according to position

- What is the structure of Golgi apparatus?
- What are the main functions of Golgi body?
- What is mitochondria? what is its main function?
- What is ribosome (structure & function)?
- From which process the endosome is produced ?